

VALIDATION REPORT

OPERATIONAL QUALIFICATION

SHELF LIFE & DEGRADATION ANALYSIS MODULE

JR Validated Environment — Shelf Life & Degradation Analysis Module v2.0

Field	Value
Document Number	JR-VR-SHELF-001
Title	Validation Report — Operational Qualification, Shelf Life & Degradation Analysis Module
Validation Plan	JR-VP-SHELF-001 v2.0
System	JR Validated Environment — Shelf Life Module
Module Version	2.0
Document Version	2.0
Execution Date	2026-05-18
OQ Result	PASS — 72 / 72 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-04-19	Joep Rous	Initial OQ execution report. All 57 test cases passed. No deviations.
2.0	2026-05-18	Joep Rous	Updated for Shelf Life module v2.0. Added --report DV report TCs for all 5 scripts (Q10-011..013, ARR-011..013, LIN-017..019, POOL-011..013, EXT-012..014). 72 / 72 tests passed.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-05-18
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Shelf Life & Degradation Analysis Module, version 2.0. It provides objective evidence that all Shelf Life scripts perform as specified in OQ Validation Plan JR-VP-SHELF-001 v2.0 under the defined test conditions.

This report records the test environment, execution results for all 72 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by admin_shelf_life_oq.

2. Scope

2.1 In Scope

- All 5 R scripts in repos/shelf_life/R/ (jrc_shelf_life_q10, jrc_shelf_life_arrhenius, jrc_shelf_life_linear, jrc_shelf_life_poolability, jrc_shelf_life_extrapolate)
- Correct computation of AF and real-time equivalent via Q10 (ASTM F1980) and Arrhenius (ISO 11607 / ICH Q1E) methods; numerical correctness verified by independent Python arithmetic
- Correct linear degradation modelling: Brown-Forsythe test, OLS regression, shelf life estimation as CI of predicted mean crossing the specification limit (ICH Q1E)
- Correct --transform log behaviour: log-scale OLS fit with CI back-transformed via exp()
- Correct ICH Q1E batch poolability: two-step ANCOVA at $\alpha=0.25$; FULL/PARTIAL/DO NOT POOL decision
- Correct extrapolation with transform-aware CI back-transformation; ICH Q1E 50% warning and 100% hard stop
- Correct rejection of invalid inputs; bypass-protection verification (RENV_PATHS_ROOT check)
- PNG and model CSV output to ~/Downloads/

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real product stability data
- Validation of R interpreter and third-party packages (ggplot2, nlme)
- Statistical validation of the underlying methods (Arrhenius kinetics, ICH Q1E poolability rules, OLS inference) — see JR-VP-SHELF-001 Section 3 for references

3. Reference Documents

Document ID	Title	Location
JR-VP-SHELF-001 v2.0	Validation Plan — OQ, Shelf Life & Degradation Analysis Module	repos/shelf_life/docs/shelf_life_validation_plan.pdf
JR-VP-001 v1.0	Validation Plan — Installation Qualification	docs/
JR-IQ-001	IQ Execution Evidence	~/jrscript/<PROJECT_ID>/validation/IQ_validation_*.txt
OQ Evidence	Shelf Life OQ Execution Evidence (pytest output + header)	shelf_life_oq_execution_20260518T172824.txt

4. Test Environment

Component	Details
Execution date / time	2026-05-18T17:28:24
Host	Joeps-MacBook-Air.local
Operating System	15.7.3
R Version	4.6.0
Python Version	3.11.9
pytest Version	pytest 8.3.4
Test runner	repos/shelf_life/admin_shelf_life_oq
Test file	repos/shelf_life/oq/test_shelf_life_suite.py
Test data	repos/shelf_life/oq/data/ (13 CSV fixtures)
Execution command	bash repos/shelf_life/admin_shelf_life_oq
OQ evidence file	shelf_life_oq_execution_20260518T172824.txt

5. Test Results

All 72 test cases were executed on 2026-05-18. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

Script	Tests Executed	Result
jrc_shelf_life_q10	13	13/13 PASS
jrc_shelf_life_arrhenius	13	13/13 PASS
jrc_shelf_life_linear	19	19/19 PASS
jrc_shelf_life_poolability	13	13/13 PASS
jrc_shelf_life_extrapolate	14	14/14 PASS
TOTAL	72	72/72 PASS

5.1 jrc_shelf_life_q10 — 13/13 PASS

TC ID	Description	Expected	Result
TC-SHELF-Q10-001	Valid inputs	exit 0; key sections in output	PASS
TC-SHELF-Q10-002	Known AF = 8.0 (± 0.001), real_time = 208.0 (± 0.01)	Independent Python arithmetic reference	PASS
TC-SHELF-Q10-003	Sensitivity table rows Q10 -0.5, Q10, Q10+0.5	exit 0; '1.5', 'stated value', '2.5' in output	PASS
TC-SHELF-Q10-004	Q10=1.2 out-of-range warning	exit 0; range warning in output	PASS
TC-SHELF-Q10-005	accel_temp $\leq$ real_temp rejected	exit non-zero	PASS
TC-SHELF-Q10-006	No arguments	exit non-zero or usage	PASS
TC-SHELF-Q10-007	Non-numeric q10	exit non-zero	PASS
TC-SHELF-Q10-008	Negative accel_time	exit non-zero	PASS
TC-SHELF-Q10-009	--help flag	exit 0; 'Usage' in output	PASS
TC-SHELF-Q10-010	Bypass protection	exit non-zero; RENV_PATHS_ROOT in output	PASS
TC-SHELF-Q10-011	--report $\rightarrow$ *_q10_dv_report.docx or .html in ~/Downloads/	exit 0; .docx or .html file found with recent mtime	PASS
TC-SHELF-Q10-012	--report JSON sidecar *_q10_dv_report_data.json (or .docx accepted)	exit 0; .docx or JSON found with recent mtime	PASS

TC-SHELF-Q10-013	--report JSON: report_type=="dv", verdict_pass is True	exit 0; if JSON present: report_type=="dv", verdict_pass is True	PASS
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5.2 jrc_shelf_life_arrhenius — 13/13 PASS

TC ID	Description	Expected	Result
TC-SHELF-ARR-001	Valid inputs	exit 0; key sections in output	PASS
TC-SHELF-ARR-002	Known AF = 13.7826 (± 0.001), real_time = 358.35 (± 0.05)	Independent math.exp() reference	PASS
TC-SHELF-ARR-003	Kelvin inputs give same AF as Celsius	AF matches ± 0.01	PASS
TC-SHELF-ARR-004	Sensitivity table Ea-2, Ea, Ea+2	exit 0; all three values in output	PASS
TC-SHELF-ARR-005	Ea=30 out-of-range warning	exit 0; range warning in output	PASS
TC-SHELF-ARR-006	accel_temp $\leq$ real_temp rejected	exit non-zero	PASS
TC-SHELF-ARR-007	No arguments	exit non-zero or usage	PASS
TC-SHELF-ARR-008	Invalid --unit F	exit non-zero	PASS
TC-SHELF-ARR-009	--help flag	exit 0; 'Usage' in output	PASS
TC-SHELF-ARR-010	Bypass protection	exit non-zero; RENV_PATHS_ROOT in output	PASS
TC-SHELF-ARR-011	--report → *_arrhenius_dv_report.docx or .html in ~/Downloads/	exit 0; .docx or .html file found with recent mtime	PASS
TC-SHELF-ARR-012	--report JSON sidecar *_arrhenius_dv_report_data.json (or .docx accepted)	exit 0; .docx or JSON found with recent mtime	PASS
TC-SHELF-ARR-013	--report JSON: report_type=="dv", verdict_pass is True	exit 0; if JSON present: report_type=="dv", verdict_pass is True	PASS

5.3 jrc_shelf_life_linear — 19/19 PASS

TC ID	Description	Expected	Result
TC-SHELF-LIN-001	Valid 15-row dataset	exit 0; key sections in output	PASS
TC-SHELF-LIN-002	Known shelf life = 24.723 ± 0.05	Independent Python OLS + bisection reference	PASS
TC-SHELF-LIN-003	Homogeneous variance — BF pass	exit 0; 'homogeneous' in output	PASS
TC-SHELF-LIN-004	Heterogeneous variance — warning, exit 0	warning in output	PASS
TC-SHELF-LIN-005	PNG saved to ~/Downloads/	*_jrc_shelf_life_linear.png found	PASS
TC-SHELF-LIN-006	Model CSV saved to ~/Downloads/	*_jrc_shelf_life_linear_model.csv found	PASS
TC-SHELF-LIN-007	--direction high	exit 0; 'upper' or 'high' in output	PASS
TC-SHELF-LIN-008	Values below spec at t=0 (direction=low)	exit non-zero	PASS
TC-SHELF-LIN-009	No arguments	exit non-zero or usage	PASS
TC-SHELF-LIN-010	Missing 'value' column	exit non-zero; 'value' in error	PASS
TC-SHELF-LIN-011	< 3 distinct time points	exit non-zero	PASS
TC-SHELF-LIN-	Bypass protection	exit non-zero; RENV_PATHS_ROOT in	PASS

012		output	
TC-SHELF-LIN-013	--transform log — happy path	exit 0; 'log' in output	PASS
TC-SHELF-LIN-014	--transform log — known shelf life (± 0.05)	Independent Python log-OLS + bisection reference	PASS
TC-SHELF-LIN-015	--transform log with value ≤ 0	exit non-zero; positive/ ≤ 0 in error	PASS
TC-SHELF-LIN-016	--transform log — model CSV transform field	model CSV contains transform = log	PASS
TC-SHELF-LIN-017	--report → *_shelf_life_linear_dv_report.docx or .html in ~/Downloads/	exit 0; .docx or .html file found with recent mtime	PASS
TC-SHELF-LIN-018	--report JSON sidecar *_shelf_life_linear_dv_report_data.json (or .docx accepted)	exit 0; .docx or JSON found with recent mtime	PASS
TC-SHELF-LIN-019	--report JSON: report_type=="dv", verdict_pass is True (homogeneous dataset)	exit 0; if JSON present: report_type=="dv", verdict_pass is True	PASS

5.4 jrc_shelf_life_poolability — 13/13 PASS

TC ID	Description	Expected	Result
TC-SHELF-POOL-001	Valid 3-batch dataset	exit 0; 'Decision', 'POOL', Step 1/2 in output	PASS
TC-SHELF-POOL-002	FULL POOL decision; both ANCOVA steps $p > 0.25$	exit 0; 'FULL POOL'; $p1 > 0.25$; $p2 > 0.25$	PASS
TC-SHELF-POOL-003	DO NOT POOL; Step 1 $p < 0.05$	exit 0; 'DO NOT POOL'; Step 1 $p < 0.05$	PASS
TC-SHELF-POOL-004	PARTIAL POOL; Step 1 $p > 0.25$, Step 2 $p < 0.01$	exit 0; 'PARTIAL POOL'; $p1 > 0.25$; $p2 < 0.01$	PASS
TC-SHELF-POOL-005	PNG saved to ~/Downloads/	*_jrc_shelf_life_poolability.png found	PASS
TC-SHELF-POOL-006	No arguments	exit non-zero or usage	PASS
TC-SHELF-POOL-007	Missing 'batch' column	exit non-zero; 'batch' in error	PASS
TC-SHELF-POOL-008	Only 1 batch	exit non-zero	PASS
TC-SHELF-POOL-009	< 3 time points per batch	exit non-zero	PASS
TC-SHELF-POOL-010	Bypass protection	exit non-zero; RENV_PATHS_ROOT in output	PASS
TC-SHELF-POOL-011	--report → *_poolability_dv_report.docx or .html in ~/Downloads/	exit 0; .docx or .html file found with recent mtime	PASS
TC-SHELF-POOL-012	--report JSON sidecar *_poolability_dv_report_data.json (or .docx accepted)	exit 0; .docx or JSON found with recent mtime	PASS
TC-SHELF-POOL-013	--report JSON: report_type=="dv", verdict_pass is True (poolable dataset)	exit 0; if JSON present: report_type=="dv", verdict_pass is True	PASS

5.5 jrc_shelf_life_extrapolate — 14/14 PASS

TC ID	Description	Expected	Result
TC-SHELF-EXT-001	Valid model CSV, target within	exit 0; 'Projection', 'Fitted value'	PASS

	range		
TC-SHELF-EXT-002	Known fit = 84.502 (± 0.01), CI lower = 83.976 (± 0.05)	Closed-form from known coefficients + NIST t_crit	PASS
TC-SHELF-EXT-003	Target within range, above spec — PASS	exit 0; 'supported' in output	PASS
TC-SHELF-EXT-004	Target beyond shelf life — FAIL message	exit 0; 'NOT supported' or 'crossed'	PASS
TC-SHELF-EXT-005	>50% extrapolation warning	exit 0; warning in output	PASS
TC-SHELF-EXT-006	>100% extrapolation hard stop	exit non-zero; 'rejected' or '100' in output	PASS
TC-SHELF-EXT-007	No arguments	exit non-zero or usage	PASS
TC-SHELF-EXT-008	Missing model file	exit non-zero; 'not found' in output	PASS
TC-SHELF-EXT-009	Wrong-script model file	exit non-zero; 'script' in error	PASS
TC-SHELF-EXT-010	Negative target_time	exit non-zero	PASS
TC-SHELF-EXT-011	Bypass protection	exit non-zero; RENV_PATHS_ROOT in output	PASS
TC-SHELF-EXT-012	--report → *_extrapolate_dv_report.docx or .html in ~/Downloads/	exit 0; .docx or .html file found with recent mtime	PASS
TC-SHELF-EXT-013	--report JSON sidecar *_extrapolate_dv_report_data.json (or .docx accepted)	exit 0; .docx or JSON found with recent mtime	PASS
TC-SHELF-EXT-014	--report JSON: report_type=="dv", verdict_pass is True (CI > spec at target=20)	exit 0; if JSON present: report_type=="dv", verdict_pass is True	PASS

6. Deviations

No deviations were identified during OQ execution. All 72 test cases passed.

7. OQ Conclusion

The Operational Qualification of the JR Validated Environment Shelf Life & Degradation Analysis Module v2.0 is complete.

Result: PASS

All 72 automated test cases specified in JR-VP-SHELF-001 v2.0 were executed on 2026-05-18 and 72 passed. Results are derived from the OQ evidence file listed below.

- jrc_shelf_life_q10 — Accelerated Ageing via Q10 Method (ASTM F1980)
- jrc_shelf_life_arrhenius — Accelerated Ageing via Arrhenius Kinetics (ISO 11607 / ICH Q1E)
- jrc_shelf_life_linear — Linear Degradation Model with Shelf Life Estimation (ICH Q1E)
- jrc_shelf_life_poolability — ICH Q1E Batch Poolability Analysis (Two-Step ANCOVA)
- jrc_shelf_life_extrapolate — CI Projection to Target Time with Transform Support

Evidence of test execution is archived at: shelf_life_oq_execution_20260518T172824.txt

The Shelf Life & Degradation Analysis module is released as version 2.0 of repos/shelf_life/. Any subsequent change to the scripts, wrappers, or help files within repos/shelf_life/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.